Primitive geometries from subgroup pairs

We treat the mathematical results in the twelve-page journal edition of Dimitri Leemans and Bernardo G. Rodrigues, *On primitive geometries of rank two*, Art Discrete Appl. Math. 2 (2019), P1.08. The original Rudvalis count is false. The previously accepted orbit bound already proved this; the present treatment supplies the exact corrected count, recovers the other eleven counts, and accounts for every finite table in the article. It also records a general algorithm correction, two exchanged subgroup labels and four exchanged pairs of type diameters.

This is a treatment of that fixed article. It does not certify the contents of the electronic representative files linked from its page 2, nor claim identity with every edition of those external datasets. The distinction between complete mathematical parametrization and file verification is made precise below. Historical running times remain attributed observations.

1 The common classification and counting argument

We first recall the elementary construction attributed by the source to Tits. Given subgroups $G_i \leq G$, take the right cosets $G_i g$ as objects of type i, with incidence defined by nonempty intersection. The base cosets G_i form a chamber since all contain identity. Right multiplication preserves types and incidence. It is transitive on objects of a given type, and on incident pairs of any two fixed types: a common element g sends the base pair to the given pair. These statements concern the incidence structure and flags of size at most two; no general higher-rank flag-transitivity assertion is being assumed. Cosets are tagged by their types, so two equal subsets with different types remain different objects.

For distinct maximal subgroups H, K in a finite group, the rank-two incidence graph is connected because $\langle H, K\rangle = G$. Its two degrees are $[H : H \cap K]$ and $[K : H \cap K]$, both at least two. Maximality makes the two coset actions primitive. Thus the source's geometries are precisely unordered pairs of distinct maximal subgroups, with equivalence defined by $\mathrm{Aut}(G)$. The distinguished group action is retained; this is not a claim about every abstract isomorphism of the underlying graphs.

Theorem 1. *Put $A = \mathrm{Aut}(G)$ and choose one representative M_i of each A-orbit of maximal subgroups of G. Let $N_i = N_A(M_i)$. A complete parameter set for the source's primitive rank-two geometries is*

$$\coprod_{i<j} N_i \backslash A / N_j \quad \sqcup \quad \coprod_i \Big((N_i \backslash A / N_i) \setminus \{N_i\} \Big) / (N_i a N_i \sim N_i a^{-1} N_i). \tag{1}$$

The parameter represented by a gives the coset geometry of $\{M_i^a, M_j\}$. Here the automorphisms act on the right, so $(M^a)^b = M^{ab}$; for an inner automorphism this is the familiar notation $M^a = a^{-1}Ma$. In general N_i means the setwise stabilizer of M_i in A.

Proof. Any pair of maximal subgroups can be written $\{M_i^x, M_j^y\}$. Conjugating by y^{-1} gives $\{M_i^{xy^{-1}}, M_j\}$. When $i \neq j$, the two distinct A-classes fix the roles of the two subgroups. Two resulting pairs are equivalent if and only if their parameters differ by left multiplication by N_i

and right multiplication by N_j. This is exactly the double-coset relation. When $i = j$, the double coset N_i gives equal subgroups and must be removed. Interchanging the two subgroups and then conjugating by a^{-1} replaces a by a^{-1}. This is the only additional identification. The preceding coset argument supplies a firm, connected primitive geometry for every remaining parameter. □

This description supplies every class and the exact equivalence relation, not merely a lower bound. It is finite and constructive once the finite group and its maximal representatives are supplied. One may list its double cosets, or use the corrected coset-orbit algorithm proved in the accompanying algorithm note. That note retains the original second subgroup and lifts only the point stabilizer. For a general group, lifting both images replaces K by $K \operatorname{core}_G(H)$; a four-element counterexample proves that the original unrestricted pseudocode is false. For the twelve simple groups in the source, the proper coset actions are faithful, so this correction does not itself change their totals.

To count (1), let Ω be the set of maximal subgroups and let $\psi(a) = |\Omega^a|$. A fixed two-element subset either consists of two fixed elements or is a two-cycle. Its number is therefore

$$\frac{\psi(a)^2 - 2\psi(a) + \psi(a^2)}{2}.$$

Burnside's lemma gives

$$\#\binom{\Omega}{2}/A = \frac{1}{2}\sum_{[a]_A} \frac{\psi(a)^2 - 2\psi(a) + \psi(a^2)}{|C_A(a)|}. \tag{2}$$

Here Ω is the disjoint union of the transitive actions A/N_i. The accompanying common proof establishes the normalizer selection and shows how complete character or mark data give every term of (2). Its product-of-marks theorem also recovers each double-coset orbit-length multiset. These are classical counting methods; their classical origin is not presented as a new discovery.

2 The twelve group-specific inputs

For $M_{11}, M_{23}, M_{24}, J_1, Ru$, the accepted character application uses complete maximal-subgroup lists, their stored character fusions, centralizers and square maps. The numbers of maximal classes are respectively $5, 7, 9, 7, 15$. Since these groups have trivial outer automorphism groups, $A = G$ and maximal subgroups are self-normalizing. Their primitive permutation characters therefore give exactly the summands of ψ. The named-group and complete-library contracts remain explicit premises.

For $M_{12}, J_2, J_3, HS, McL, He$, the acquired complete subgroup tables of $A = G.2$ identify the socle G, its maximal subgroups up to A, and their actual normalizer classes in A. The resulting numbers of maximal A-classes are $8, 9, 8, 10, 10, 10$. Their normalizer mark rows, evaluated at the stored cyclic-class fusion, give ψ on every element class of A, including the outer classes. Thus neither (1) nor (2) omits outer identifications. The exact maximal and normalizer class lists are retained in the group-specific input map, with their table editions.

For M_{22}, the previously accepted L073 census supplies the complete $\mathrm{Aut}(M_{22})$ action and its 81 rank-two classes. Its representative construction and completeness proof are imported, not rerun. The separate simple-group mark calculation gives 104 inner classes, which must not be substituted for the required full-automorphism total.

These inputs specify every family in (1), while (2) computes its exact cardinality. They do not assert that permutation characters alone distinguish all embeddings: the two degree-253 maximal subgroups of M_{23} are a concrete exception, resolved by the additional normal-subgroup certificate. The explicit M_{11} calculation further constructs all 37 representatives, their coset graphs and their design data.

Group	printed total	recovered total	input route
M_{11}	37	37	characters and literal pairs
M_{12}	166	166	marks of $M_{12}.2$
M_{22}	81	81	accepted L073 census
M_{23}	170	170	characters
M_{24}	5074	5074	characters
J_1	669	669	characters
J_2	334	334	marks of $J_2.2$
J_3	546	546	marks of $J_3.2$
HS	339	339	marks of $HS.2$
McL	443	443	marks of $McL.2$
He	4074	4074	marks of $He.2$
Ru	1511	556287	characters

The earlier Rudvalis lower bound 50669 already disproved 1511. The exact value 556287 completes that correction; it is not a second principal refutation. Summing the corrected table gives 568220 classes across the twelve groups, under the specified group equivalences.

The groups' standard orders and the degrees of the source's chosen permutation representations are retained as named-group data. The displayed runtimes concern the authors' implementation and workstation. Our exact counting proof does not reproduce those benchmarks or certify a speedup.

3 The finite tables and the correction ledger

The product-of-marks calculation recovers all 308 numerical cells in Tables 2 and 5–8. In Table 7 the two degree-253 M23 row labels are interchanged. The acquired class of order 40320 with orbits $7, 16$ on the 23 points contains a unique conjugate of an elementary abelian subgroup of order 16, and is $2^4 : A_7$. The other is $L_3(4) : 2$. The detailed proof counts contained conjugates by $m_{M,L}|M|/|N_G(L)|$ and uses the complete cyclic fusion to prove exponent two. No numeric table cell changes.

The preliminary upper-triangle counts 38 for M11 and 268 for M12 are also recovered. Identifying diagonal double cosets under inversion gives the inner counts 37 and 255; the full M12 automorphism action then gives 166. The three quantities have different meanings and are not compared as if they were the same enumeration.

For Tables 3 and 4, the concrete M11 certificate covers 309 maximal subgroups and all 47586 unordered pairs. It obtains 37 orbits, checks all coset graphs and transports every named intersection to an accepted structural proof. All 70 design parameters agree. With the printed subgroup columns retained, four type-diameter triples change:

source rows	printed	corrected
$12, 18$	$[3, 2, 4]$	$[4, 2, 3]$
$23, 35$	$[3, 3, 4]$	$[4, 3, 3]$.

The definition of d_p and d_l is in *Section 2, physical page 3*, before Section 3 begins. The frozen M11 application mistakenly called this Section 3. The present synthesis corrects that citation only; its source page, four numerical corrections and proofs are unchanged. The publication edition of the M11 text records the same explicit citation delta.

The M11 note gives a direct witness for row 35: the graph is the subdivision of K_{11}, with eccentricity four at an edge vertex and three at an original vertex. The two subgroup indices in the printed order are 55 and 11. This explains the correction independently of the graph program. Rows 10/11 and 16/17 have indistinguishable printed profiles; both possible matches are retained, while their actual pair orbits remain distinct.

The source omits design rows 31 and 32. They are complete incidence with repeated full blocks. They meet its literal subset-counting definition, but not a convention requiring proper or distinct blocks. We retain that qualification explicitly. It does not change any of the seventy printed design parameters. The undefined K in the pseudocode is a separate variable typo; it is distinct from the substantive core-kernel issue.

4 Precisely what is, and is not, completed

Theorem 1, the corrected finite enumerator, the acquired complete group-specific inputs and the exact orbit counts provide a complete mathematical parametrization for all twelve groups. The individual finite invariants printed in the paper are recovered with the corrections just listed. The source's comparison to earlier M11, M22 and M23 totals is accounted for by these same exact counts and the L073 import. We do not expand (1) into a file listing all 568220 representatives. The classification supplied here is the explicit parameter set, its equivalence relation and its exact size, with all the finite invariants actually printed in the article checked separately.

The external data addresses on source page 2 are `data.adam-journal.eu/apg/` and the authors' former `homepages.ulb.ac.be/~dleemans/PRI/` directory. Their downloadable representative files have not been read class by class, matched against our parameters, or certified complete by this treatment. In particular, our exact Ru count does not validate any external file carrying the old 1511 total. Those files are separate data artifacts, not extra input premises silently used to prove the article's finite tables.

The Tits construction and the earlier Dehon algorithm are attributed background results. The source's McLaughlin-code examples refer to another paper; they are not newly proved code theorems here. Its suggestion to extend the new algorithm to higher ranks is explicitly future work, and its explanation of a possible earlier M22 error is speculation. Neither is converted into a theorem or a solved-conjecture claim.

Our contribution is the common classification and counting treatment, its applications with explicit input contracts, the finite certificates, and the stated corrections. Burnside's lemma, tables of marks, transporter counting and coset geometry are borrowed mathematics. The structural classifications and library completeness used in the applications remain visible premises. This source treatment claims no new ordinary recovery, second principal refutation, Lean formalization or compression factor. Acceptance of the complete corrected disposition is a separate review gate.